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Manipulating charge transfer dynamics and stabilizing lead bromide octahedra for efficient blue perovskite light-emitting diodes

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Blue light-emitting diodes based on reduced-dimensional pure bromide perovskites have attracted increasing attention due to their superiority in achieving stable and efficient blue emissions. Yet, the employment of abundant insulating long-chain organic spacer cations severely limits the charge carriers transport capability and electroluminescence performance. Here, we utilize short-chain iminodi(methylphosphonic acid) ligand with two-terminated phosphonic groups that can strongly interact with the lead bromide octahedra via synergetic covalent bonds, facilitating the interlayer charge transfer and simultaneously reduces the defect density in perovskite films, thereby improving the light emission efficiency. Iminodi(methylphosphonic acid) ligand also enhances the dielectric confinement, leading to higher exciton binding energy and accelerated radiative recombination in perovskite emitters. The resulting blue light-emitting diode achieves a maximum external quantum efficiency of 25.4% and an operational lifetime exceeding 800 min, which represents a significant advancement among state-of-the-art blue light-emitting diodes based on pure bromide perovskites.

Perovskite light-emitting diodes (PeLEDs) are promising candidates for next-generation full-color displays and solid-state lighting due to their narrow emission linewidth, high efficiency, wide color gamut and solution processability[1–4]. Significant progress has been achieved in red and green PeLEDs, which presents external quantum efficiencies (EQEs) exceeding 25% and extended operational stability surpassing 100,000 hours[5–8]. However, the development of efficient and stable blue PeLEDs remains a substantial challenge[9,10], impeding the practical applications and the commercialization of perovskite-based display technology. At present, incorporating chlorine element into pure-bromide (Br) perovskite emitters is one of the most widely adopted strategies for tuning the emission wavelengths from green to blue region[11–13], but suffers badly from the ion migration resulting in the spectra shifting under electric field and device instability[14]. Besides, a high chlorine ratio required for bandgap extension generally contributes to the generation of deep-level defect states and the

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decreased film quality, which together deteriorate the charge transport and the radiative recombination of excitons in PeLED[15–17]. The halide segregation can be effectively circumvented in small colloidal pure Br perovskite quantum dots (QDs, 3–5 nm) with strong quantum size effect, whereas the synthesis of uniform and monodispersed sub-5 nm-sized QDs is rather complicated, and the long-chain organic capping ligands detrimentally impede efficient charge carrier transport in PeLEDs[18].

Pure-Br reduced-dimensional perovskites with quantum well structure have been identified as promising blue emitters, due to their higher exciton binding energy and enhanced moisture resistance as compared to their 3D bulk counterparts[19,20]. So far, researchers have mainly focused on either modulating the quantum confinement effect in pure-Br quasi-2D perovskite via spacer engineering to tune their bandgaps in the entire blue region[21], or passivating the crystal defects and the phase transition via additive engineering and/or dielectric engineering to improve the radiative recombination efficiency of excitons in PeLEDs[22,23]. Unfortunately, the reported blue LED based on pure-Br perovskite presents an EQE of ~18%[24], far behind the state of the art. The further enhancement of their electroluminescence (EL) performance is still hindered by the inferior charge transport/ injection capability and the deep-level defects induced nonradiative loss of excitons. It is significant to explore a comprehensive avenue to fundamentally resolve the dilemma from the root.

Here, we realize the remarkable performance enhancement in blue LEDs based on pure-Br perovskites enabled by employing a short-chain ligand of iminodi(methylphosphonic acid) (IDMP) with two-terminated phosphonic groups, which have strong electron-withdrawing ability and can coordinate with two adjacent lead bromide ($[PbBr_6]^{4-}$) octahedra via synergetic covalent bonds (Pb-O-P). Moreover, the IDMP ligand also enhances the dielectric confinement in perovskite films, resulting in higher exciton binding energy and accelerated radiative recombination. The as-prepared perovskite film presents both higher photoluminescence quantum yield (PLQY) and higher conductivity. Notably, the resistance ability of the IDMP-treated film to external conditions including electric field, thermal heating, irradiation exposure, etc., is also remarkably enhanced. As a result, we demonstrate efficient and stable blue LEDs using the IDMP-treated perovskites as the emissive layer. The optimized device presents a peak EQE of 25.4% at 487 nm and maximum luminance of 2459 cd m^{-2}, which are nearly two-fold higher than those for control device based on pristine film. The operational stability of the optimized device shows a half-lifetime (T_{50}) of 845 min at initial luminance of 100 cd m^{-2}, which is nearly nine times longer than that for control device (95 min).

Results

Regulating the exciton dynamics in perovskites

We introduce a short-chain ligand, namely iminodi(methylphosphonic acid) (IDMP), into the precursor solution to in situ interact with the pure-Br perovskite with a composition of $(PBA/MOPA)_2Cs_{n-1}Pb_nBr_{3n+1}$ during the crystallization stage (see Methods for details). Figure 1a shows the molecular structure and electrostatic potential (ESP) of the IDMP ligand, consisting of a centered amino (-NH-) group and two terminated phosphonic acid ($-CH_2PO_3H_2$) groups with strong electron-withdrawing ability. The half-IDMP-structured ligand, namely (aminomethyl)phosphonic acid (AMP, Supplementary Fig. 1) is also investigated as a comparison. We perform capacitance-voltage (*C-V*) measurements to evaluate the average dielectric constant (ε) of pristine, AMP-treated, and IDMP-treated perovskite films, which are calculated to be 8.0, 7.3 and 6.5, respectively (Fig. 1b, Supplementary Fig. 2). The reduced ε is expected to enhance the conductivity and induce the increase in the dielectric confinement in quasi-two-dimensional (quasi-2D) perovskite[25]. As derived from the temperature-dependent PL spectra results, the IDMP-treated film shows the highest exciton binding energy (E_b) of 111.3 meV, as compared with those for pristine (77.8 meV) and AMP-treated (88.4 meV) films (Fig. 1b, Supplementary Fig. 3). Notably, the optical bandgap (E_g) for both AMP- and IDMP-treated samples remains unchanged (Supplementary Fig. 4). These results imply that the AMP and IDMP ligands hardly affect the quantum confinement effect but

We then investigate the effects of the AMP and IDMP ligands on the optical properties of perovskite films. All the perovskite films exhibit efficient PL emissions peak at approximately 487 nm and the PL intensity of the perovskite films is remarkably enhanced for the IDMP-treated sample (Fig. 1d). The corresponding exciton recombination dynamics is analyzed by time-resolved PL (TRPL) decay curves (Fig. 1e), which are fitted using a three-exponential rate law and the detailed parameters are summarized in Supplementary Table 1. The average lifetime (τ_{avg}) is prolonged from 10.89 ns for pristine film to 13.62 ns and 17.51 ns for AMP- and IDMP-treated films, respectively, suggesting a substantial reduction in trap-assisted nonradiative recombination within the IDMP-treated perovskite films[26,27]. The PLQY values for AMP- and IDMP-treated perovskite films are enhanced to 39% and 70%, respectively (Fig. 1f). The enhanced radiative (k_r) and the reduced nonradiative (k_{nr}) recombination rate constants (Supplementary Table 2) further manifest that IDMP ligand promoted the radiative recombination in perovskite effectively. Based on these above theoretical and experimental results, we propose the roles and the underlying working mechanisms of IDMP ligand at the interlayer and intra of quasi-2D perovskites, as schematic illustrated in Fig. 1g. Specifically, the $P=O$ groups from two-terminated phosphonic acids of IDMP ligand can synergistically interact with two adjacent $[PbBr_6]^{4-}$ octahedra by forming covalent bonds (Pb-O-P), which are favorable for eliminating defect states, fastening the perovskite lattice and thus, mitigating the ion migration under external conditions. More importantly, this synergistic connection modes of IDMP ligand provide efficient pathways for the interlayer charge transfer, which is highly desired for achieving efficient PeLEDs. Due to its dielectric feature, the IDMP ligand strengthens the Coulomb attraction force of excitons by enhancing the dielectric confinement effect in quasi-2D perovskite and therefore, promoting the radiative recombination efficiency.

Chemical states and carrier recombination dynamics

X-ray photoelectron spectroscopy (XPS) and Fourier transform infrared spectroscopy (FTIR) were employed to probe the interactions between IDMP ligands and perovskite lattice. The XPS results reveal that both Pb *4f* and Br *3d* peaks shift toward lower binding energy by approximately 0.2 eV after IDMP treatment (Fig. 2a–b), suggesting the existence of strong interactions between IDMP ligands and perovskite lattice[28]. Specifically, the functional $P=O$ moiety from IDMP ligands donate its lone electron pair on the oxygen atom to the empty *6p* orbital of Pb^{2+} ions, which reduces the electron cloud density of Pb^{2+}

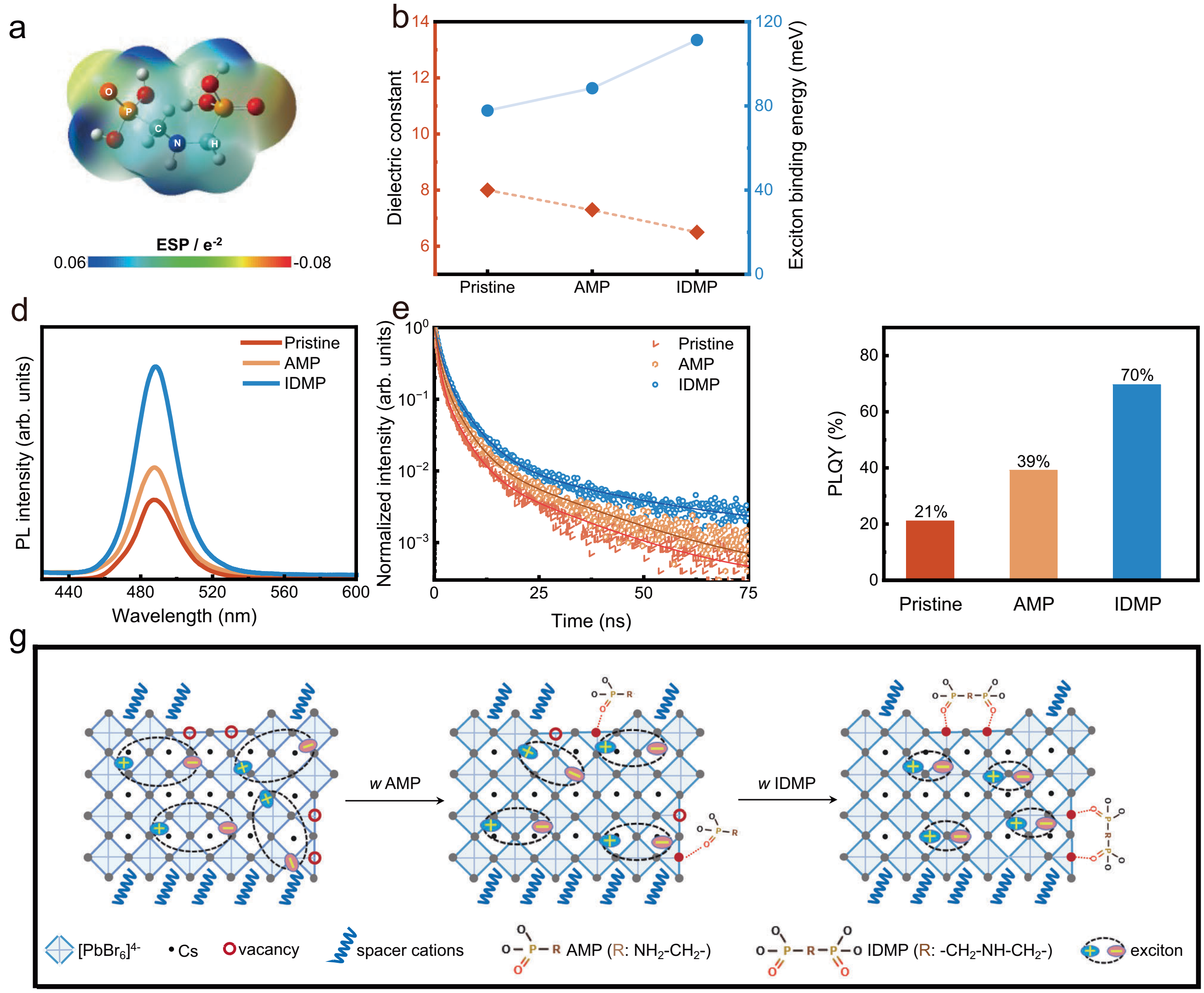

Fig. 1 | Regulation of exciton behaviors at interlayer and intra. **a** Molecular structure and calculated electrostatic surface potential map of iminodi(methylphosphonic acid) (IDMP) ligand. **b** The dielectric constant (left axis) and the corresponding exciton binding energy (right axis) for pristine, (aminomethyl) phosphonic acid (AMP) -treated, and IDMP-treated perovskite films, respectively. AMP-treated, and IDMP-treated perovskite films, respectively. **g** Schematic illustration of the working mechanism of the IDMP ligand.

and thus, affects the electrostatic interaction between the Pb-Br bonds in $PbBr_6$ octahedra[29]. In the FTIR spectra (Fig. 2c), it is observed that the peak of $P=O$ stretching vibration (ν ($P=O$), 1264 cm^{-1}) of IDMP shifts to a lower wavenumber (1260 cm^{-1}) in the presence of $PbBr_2$, which further proves the coordination interaction between IDMP and $PbBr_6$ octahedra[30]. In addition, the interaction between AMP molecules and perovskite structure is also investigated, which is weaker than that in the IDMP-treated film (Supplementary Figs. 7–8). This is also verified by the nuclear magnetic resonance (NMR) spectra of $PbBr_2$, AMP•$PbBr_2$ and IDMP•$PbBr_2$ (Supplementary Fig. 9), where the distinct upfield-shifting of the ^{207}Pb signal signifies the strong affinity of IDMP ligands[31]. The stretching vibration peak (O-P-O stretching, located at 872 cm^{-1}) of the AMP bond in $PbBr_2$ does not shift compared with pure AMP. The same phenomenon can also be observed in IDMP•$PbBr_2$, suggesting that there is no interaction between the O-P-O in AMP or IDMP and the $PbBr_2$ (Supplementary Fig. 10).

To investigate the effects of IDMP ligands on the crystallization kinetics of perovskite films, we employ in situ PL measurements to monitor the evolution of PL intensity during the dynamic spin-coating (the nucleation stage) and annealing treatment (the growth stage) processes of perovskite films. With IDMP ligands, the blue emission peak is observed before the dropping of antisolvent at 27 s and the PL intensity is substantially enhanced across the annealing stage of perovskite films, while the blue emission is not seen until the antisolvent dropping for the control perovskite film without IDMP and the corresponding PL intensity started to fade gradually since 150 s during annealing stage (Fig. 2d and Supplementary Fig. 11). We propose that the IDMP ligands facilitate the nucleation kinetic of perovskite films by enhancing the nuclei density and improve the structure integrity of perovskite due to its strong binding affinity[32]. The resulting IDMP-treated perovskite films exhibit higher crystallinity, as indicated in the XRD patterns (Supplementary Fig. 12). Scanning electron microscopy (SEM) and atomic force microscopy (AFM) analysis further reveal a smoother morphology for the IDMP-treated film, for which the surface roughness reduced to 1.70 nm from 3.11 nm for the pristine film (Supplementary Fig. 13).

We analyze transient absorption (TA) spectra to understand the effects of IDMP ligand on the photocarrier recombination dynamics in

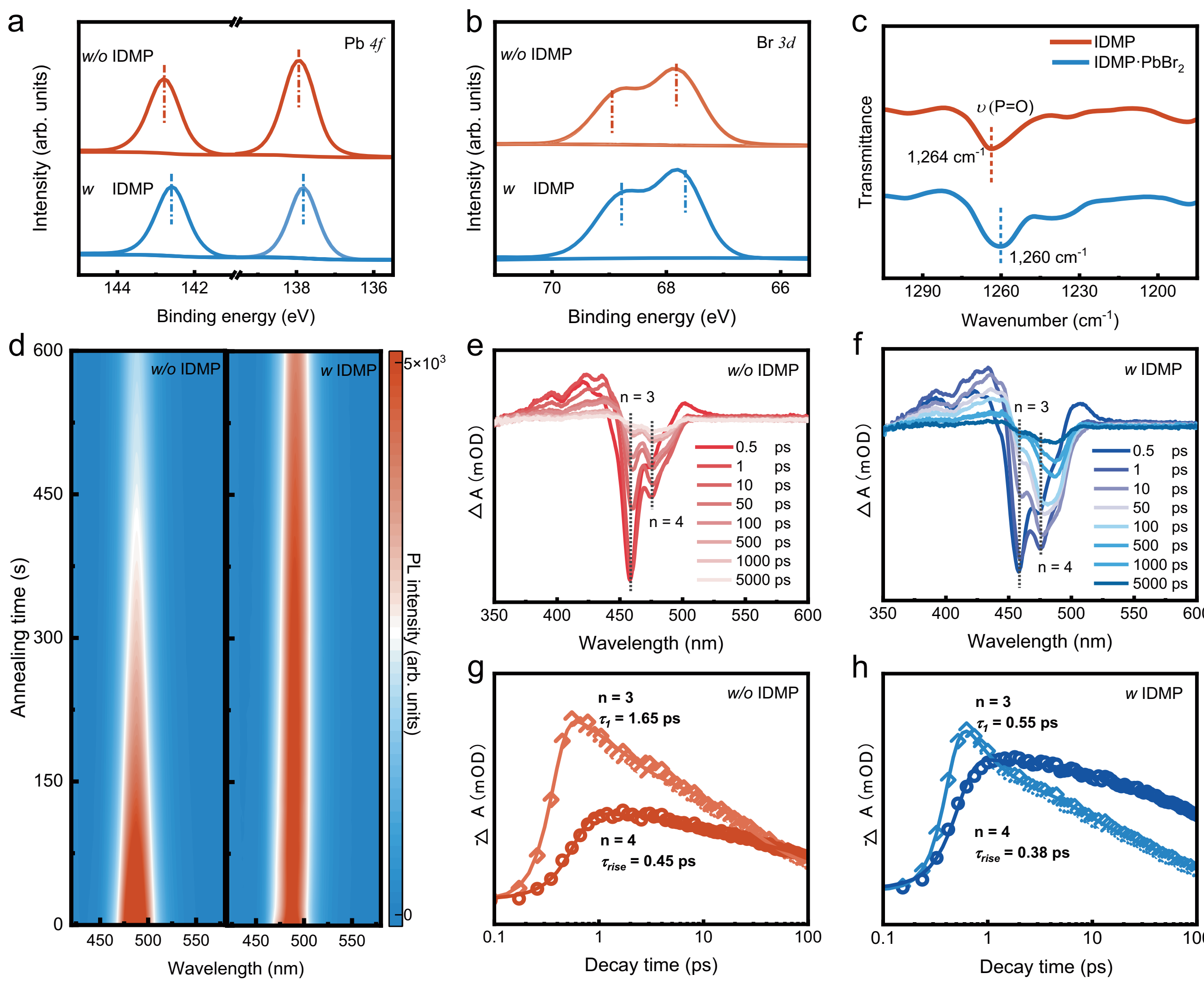

Fig. 2 | Interactions between IDMP ligands and perovskites. a, **b** X-ray photoelectron spectroscopy spectra of Pb *4f*, and Br *3d* signals for pristine and IDMP-treated perovskite films, respectively. **c** Fourier transform infrared spectroscopy spectra of pristine IDMP and IDMP•$PbBr_2$ mixture. **d** In situ photoluminescence spectra for pristine and IDMP-treated perovskite films during annealing treatment at 60 °C. **e**, **f** Transient absorption (TA) spectra with different time scales of pristine and IDMP-modified perovskite films. **g**, **h** Transient absorption kinetics are probed at selected wavelengths corresponding to $n = 3$ and $n = 4$ for pristine and IDMP-treated perovskite films.

perovskite films. Two distinct ground-state bleach (GSB) peaks at 458 nm and 477 nm, corresponding to $n = 3$ and $n = 4$ phases of quasi-2D perovskites, are observed for both films (Fig. 2e–f and Supplementary Fig. 14), which is in good agreement with the exciton absorption peaks in the steady-state absorption spectra (Supplementary Fig. 4). The photoexcited carriers are initially generated in $n = 3$ phase and then transferred to $n = 4$ phase for radiative recombination[33]. It clearly shows that the excitons are vastly accumulated in the $n = 3$ phase for pristine film with a long excitation time (~1000 ps), indicating the incomplete energy transfer between the different phases[29]. By contrast, the exciton resonance at $GSB_{n=3}$ disappeared after ~100 ps for the IDMP-treated film and refers to a more efficient energy transfer process and an enhanced interlayer charge transfer capability. We further extract the time traces of $n = 3$ and $n = 4$ phases to investigate the dynamics of the carrier transfer processes (Fig. 2g–h and Supplementary Table 3). The $n = 3$ phase exhibits a faster decay time (τ_1) of 0.55 ps and the $n = 4$ phase has a faster formation time (τ_{rise}) of 0.38 ps for IDMP-treated film, as compared to those for pristine film (τ_1 of 1.65 ps, τ_{rise} of 0.45 ps). This visually demonstrates that the IDMP ligands have significantly facilitated the energy transfer efficiency in perovskite films.

Analysis of electrical properties of perovskite films

The variation of the surface potential for perovskite films is characterized by Kelvin probe force microscope (KPFM) (Fig. 3a–b), the IDMP-treated sample displays a lower average surface potential (396 mV) as compared to that for pristine film (413 mV), which facilitates the injection/transport of holes in PeLEDs[34]. Moreover, the incorporation of IDMP also improves the homogeneity of the surface potential with a narrower distribution (Fig. 3c), indicative of uniform chemical compositions of perovskites with fewer defect states[35,36]. Hall effect measurements are performed to characterize the polarity and concentration of carriers in perovskite films[37,38]. The majority carriers are electrons with a concentration of $5.7 \pm 0.4 \times 10^{16}\ m^{-3}$ for pristine film and holes with a concentration of $1.9 \pm 0.2 \times 10^{16}\ m^{-3}$ for IDMP-treated film, respectively (Supplementary Fig. 15), implying a better hole transport capability. We also perform the space-charge-limited current (SCLC) measurements to evaluate the carrier mobility for pristine and

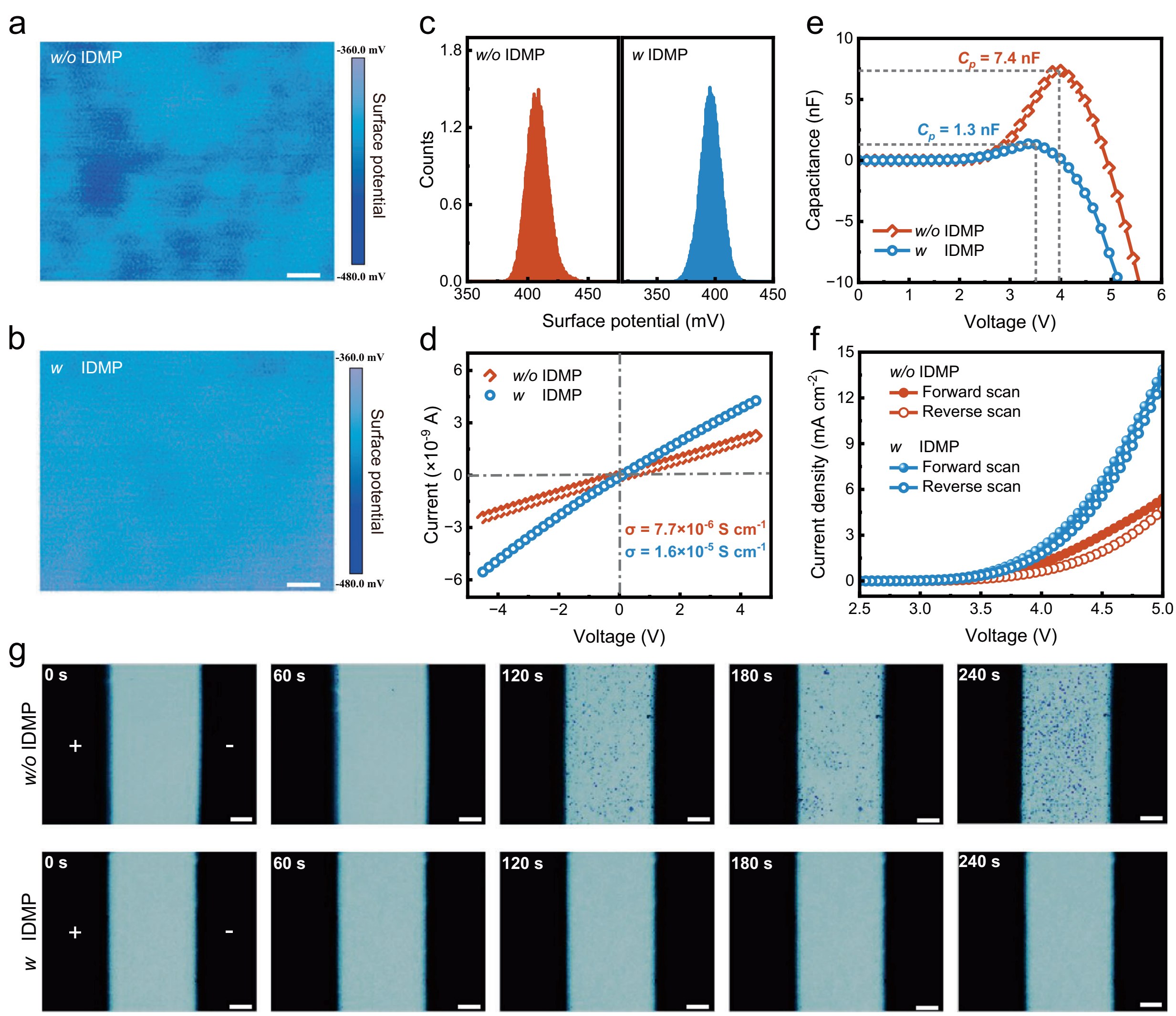

Fig. 3 | Electrical characterizations of perovskite films. a, **b** Kelvin probe force microscope images of pristine and IDMP-treated perovskite films, scale bars, 1 μm. **c** The corresponding statistics of surface potential distributions. **d** Conductivity measurements of perovskite films. **e** Capacitor-voltage characteristics, and **f** hysteresis current-voltage curves of PeLEDs based on pristine and IDMP-treated perovskite films. **g** Time-dependent microscopic images of pristine (top panel) and IDMP-treated (bottom panel) perovskite films under an external electric field (~6×10^4 V m^{-1}) using a wide-field PL imaging microscopy, scale bars, 20 μm. The bilateral black areas refer to Ag electrodes and the external electric field is marked by the "+" and "-" symbols.

IDMP-treated perovskite films, which is calculated to be 6.53×10^{-5} cm^2 V^{-1} s^{-1} and 9.41×10^{-5} cm^2 V^{-1} s^{-1}, respectively (Supplementary Fig. 16a). The conductivity of IDMP-treated perovskite film is increased by one order of magnitudes due to the increased carrier mobility (Fig. 3d), which is desired for accelerating hole injection and thus, achieving high-performance blue PeLEDs.

We then fabricate PeLEDs with a structure of indium tin oxide (ITO)/poly(3,4-ethylenedioxythiophene)-poly-(styrenesulfonate) (PEDOT:PSS)/poly(9-vinylcarbazole): 2-(4-biphenyl)-5-(4-tert -butylphenyl)-1,3,4-oxadiazole (PVK:PBD)/[2-(9H-Carbazol-9-yl) ethyl] phosphonic acid (2PACz)/perovskites film/tri[3-(3-pyridyl) mesityl] borane (3TPYMB)/lithium fluoride (LiF)/ Aluminum (Al) (Supplementary Fig. 17) and characterized the *C-V* curves (Fig. 3e). The lower voltage of peak capacitance (V_p) of PeLED based on IDMP-treated film, denoted as target device in the following part, implies the accelerated charge dynamic processes, as compared to those in the control device based on pristine film[39]. The smaller peak capacitance (C_p) value in target device further demonstrates that the interfacial charge accumulation induced by inefficient charge injection/transport is effectively suppressed[40,41], consistent with the reduced defect density in IDMP-treated film (Supplementary Fig. 16b–c). The slower decay rate of the peak capacitance in target device also implies better operational stability (Supplementary Fig. 18). The forward and reverse current density-voltage (*J-V*) curves of the corresponding devices are also collected (Fig. 3f). As seen, the forward/reverse scans in target device are nearly closely coincident due to the passivation of defect states and the suppression of ion migration. Time-dependent microscopic imaging is utilized to visualize the ion migration in perovskite films between two in-plane Ag electrodes (with a 100 μm spacing) under an applied electric field (~6×10^4 V m^{-1}) (Fig. 3g and Supplementary Fig. 19). The pristine film and AMP-treated film show severe PL degradation and spatial phase segregation after 120 s and 180 s, respectively, while the IDMP-treated film maintain stable PL emission without evident phase segregation until 240 s[42]. We further study the PL stability of perovskite films under heating and ultraviolet (UV) light exposure conditions. The PL intensity of pristine film decrease by 50% at 120 °C and almost completely diminish at 150 °C, whereas the IDMP-treated film retains 75% of its initial PL intensity at 150 °C. After 240 min of UV irradiation, the pristine film exhibits negligible PL emission, while the IDMP-treated film maintains 60% of its initial PL intensity,

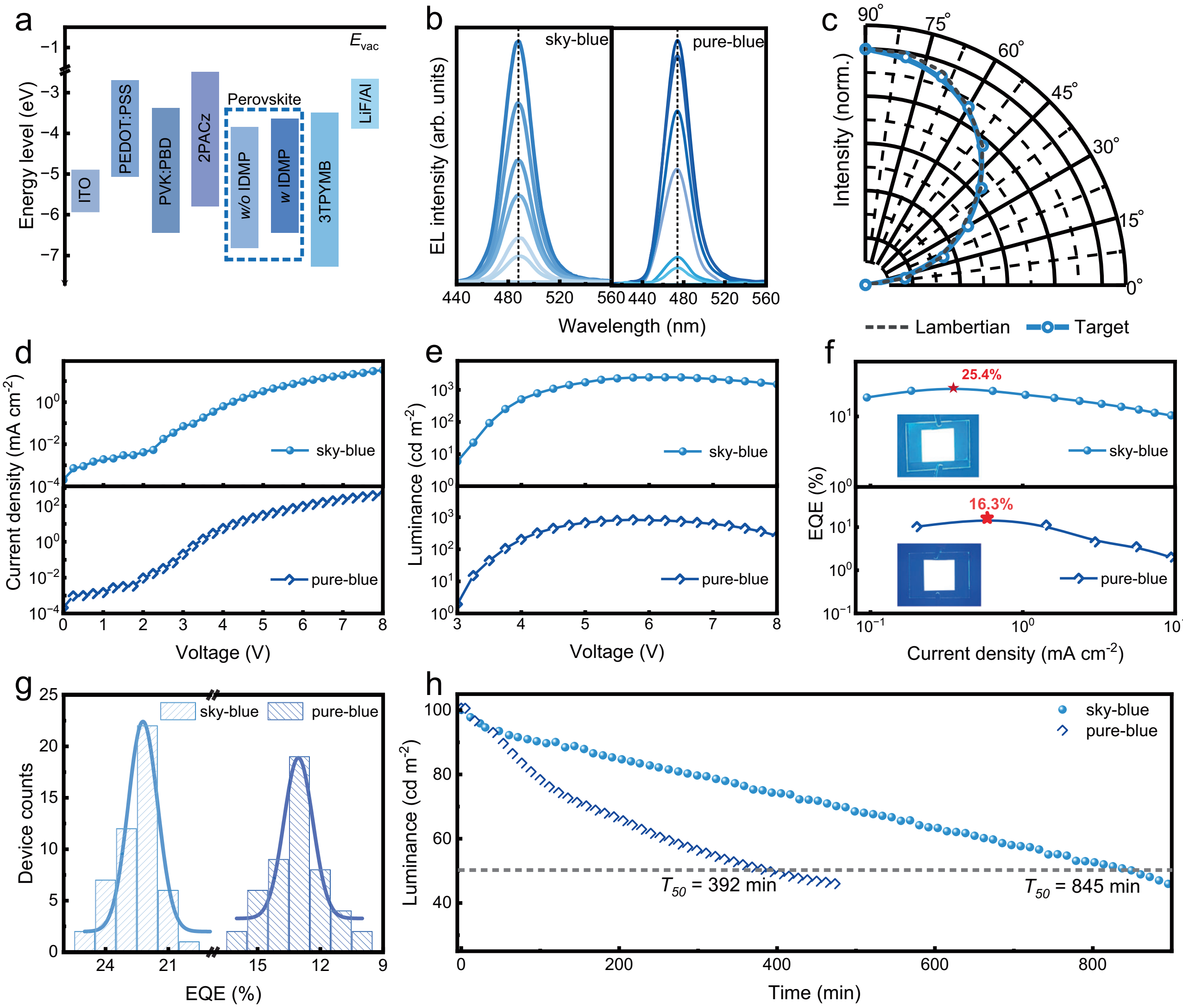

Fig. 4 | Device performance of the blue devices. a The flat band energy level diagram of the PeLEDs. **b** Electroluminescence spectra for devices under varying voltages. **c** Angle-dependent electroluminescence intensity for the sky-blue device. **d** Current density-voltage, **e** luminance-voltage, **f** external quantum efficiency-current density curves. The active area of the device for performance testing is 2×2 mm^2. The inset photographs in **f** display operating target devices (20×20 mm^2) under 4 V. **g** Histogram statistics of peak external quantum efficiencies for sky-blue and pure-blue devices, and **h** the comparison for their corresponding operational lifetime.

indicating remarkable high-temperature and UV-light stability (Supplementary Fig. 20). Based on these comprehensive results, it is plausible to conclude that the IDMP-treatment can improve the charge transfer kinetics in PeLEDs, passivate defects and suppress ion migration in perovskite films.

Electroluminescence performance and operational stability

Figure 4a illustrates the device energy-level alignment, with perovskite levels derived from ultraviolet photoelectron spectroscopy (UPS) and ultraviolet-visible absorption (Supplementary Fig. 21), and other energy level values from previous literatures[43,44]. Notably, the target device based on IDMP-treated film exhibits a reduced hole-injection barrier (0.52 eV) compared to the control device (0.65 eV). This lower barrier is expected to improve charge balance, thereby enhancing radiative recombination efficiency in the perovskite layer[45]. We obtain both sky-blue (λ_{em} = 487 nm) and pure-blue (λ_{em} = 474 nm) devices by adjusting the proportions of spacer cations of PBA^+ and $MOPA^+$ accordingly (Supplementary Fig. 22). As increasing the voltages from 4 to 9 V, the target devices show gradually enhanced EL intensity without broadening and peak shift (Fig. 4b), for which the EL intensity is higher to that of control device at the same voltage (Supplementary Figs. 23–24). The angular emission intensity of the target PeLED follows a Lambertian profile, showing a uniform distribution of light intensity across different angles, which is highly desirable for lighting and display applications (Fig. 4c). The target devices exhibit an increased current density (Fig. 4d) and higher brightness (Fig. 4e) compared to control devices--the maximum luminance of target PeLEDs (2459 cd m^{-2} and 826 cd m^{-2}) are much higher than that of control devices (1373 cd m^{-2} and 596 cd m^{-2}), and meanwhile the peak EQEs for the target devices reach 25.4% for sky-blue and 16.3% for pure-blue, respectively (Fig. 4f), approximately twofold and trifold higher than that for control devices (13.6% and 5.2%). The target sky-blue PeLEDs show an average EQE of 22.2 ± 0.13%, twofold higher than that of control ones (10.9 ± 0.11%), and the pure-blue devices also present a relatively high average EQE of 13.1 ± 0.11% (Fig. 4g). Figure 4h compare the operational stability of the PeLEDs, which are tested in a glove box (25 °C) filled with nitrogen without encapsulation. When the initial luminance is set as 100 cd m^{-2}, the operating half-lifetime (T_{50}) for our sky-blue and pure-blue device was up to 845 min and 392 min, which is nearly 9-fold and 11-fold higher than that for control devices (95 min and 35 min, respectively). We also monitor the real-time temperature of operating sky-blue target device using an infrared thermal camera. As the luminance increases from 500 to 1000 cd m^{-2}, the temperature is increased from 26.2 °C to 28.1 °C for the target PeLED and from 29.4 °C to 35.3 °C for the control device, respectively (Supplementary Fig. 25), which manifests the reduced Joule heat in target device and thus, favors the long-term operational stability. Our method is also validated in fabricating deep-blue PeLED with a maximum EQE of 3.0% at 468 nm, which is 1.7-fold than that for control device (Supplementary Fig. 26). All performance parameters are summarized in Supplementary Table 4. Our optimized target pure-blue and sky-blue devices, to the best of our knowledge, represent the record performance for blue PeLEDs based on pure-Br perovskites, and to highlight our results, pure-Br perovskite based blue LEDs, the performance of previously reported PeLEDs based on both pure-Br and mixed halide perovskites are summarized in Supplementary Table 5.

Discussion

We demonstrate the high-performance blue PeLEDs based on pure-Br perovskites by manipulating the charge transfer dynamics in layered perovskites using short chain, multi-functional IDMP intercalating ligands, which are in situ incorporated into the framework of $(PBA/MOPA)_2Cs_{n-1}Pb_nBr_{3n+1}$ quasi-2D perovskites via the formation of double-anchored covalent bonds. Benefiting from the strong interaction between IDMP and $[PbBr_6]^{4-}$ octahedra, the structural stability of the perovskite lattice under external conditions (electric field, heat, light) is significantly enhanced, and the uncoordinated Pb^{2+} ions are passivated effectively. The decrease in the average dielectric constant of the resultant IDMP-treated quasi-2D perovskite films results in the higher conductivity and larger exciton binding energy. We are thus able to realize the champion performance of blue PeLED based on pure-Br perovskite films, presenting a peak EQE of 25.4% and operational lifetime surpassing 800 min. This work broadens the pathway for developing high-performance LEDs and other optoelectronic devices based on quasi-2D perovskites.

Methods

Materials

Cesium bromide (CsBr, 99.999%) and lead bromide ($PbBr_2$, 99.999%) are purchased from Alfa Aesar. Benzenebutanammonium bromide (PBABr) is purchased from Xi'an E-Light New Material CO., Ltd. Poly(9-vinylcarbazole) (PVK), DL-α-Methoxyphenylacetic acid (MOPA), iminodi (methylphosphonic acid) (IDMP) and dimethyl sulfoxide (DMSO, 99.9%) are purchased from Sigma-Aldrich. Tri[3-(3-pyridyl)mesityl] borane (3TPYMB), and lithium fluoride (LiF) are purchased from Luminescence Technology Corp. Chlorobenzene (CB) are purchased from Sinopharm Chemical Reagent Co., Ltd. Guanidine Hydrobromide (GABr), 2-(4-biphenyl)-5-(4-tert -butylphenyl)-1,3,4-oxadiazole (PBD), poly(3,4-ethylenedioxythiophene)-poly-(styrenesulfonate) (PEDOT:PSS) are purchased Xi'an Polymer Light Technology Corp. [2-(9H-Carbazol-9-yl) ethyl]phosphonic acid (2PACz) is purchased from Aladdin. Hydrobromic acid (HBr) and absolute ethyl alcohol are purchased from Sinopharm Chemical Reagent Co., Ltd. (Aminomethyl) phosphonic acid (AMP) is purchased from Innochem. All materials are without further purification.

Perovskite precursors

CsBr, $PbBr_2$, PBABr, MOPABr, and GABr are dissolved in DMSO with the molar ratio of 1:1.2:0.8:0.4:0.2, 1:1.2:1.2:0.4:0.2, and 1:1.2:1.4:0.4:0.2 for the perovskite films with emission peak positions of 487, 474 nm and 468 nm, respectively, giving a concentration of 100 mg mL^{-1}. Then the mixture is stirred at 60 °C for night and filtered through a 0.22 μm polyterafluoroethylene membrane before using.

Device fabrication

The ITO glasses as anode are sonicated for 15 min with dish soap, deionized water, acetone and isopropyl alcohol in turn, and then dried under air flow, followed by the oxygen plasma treatment for 25 min before being transferred into a nitrogen glovebox (both H_2O and $O_2 \leq 0.1$ ppm). PEDOT:PSS is spin-coated on the ITO glass-substrate at 6,000 rpm for 40 s followed by annealing process at 150 °C for 20 min in air. For the deposition of PVK:PBD layer, PVK:PBD/CB (6 mg mL^{-1}, PVK:PBD with a mass ratio of 1:1) mixture solution is spin-coated on the PEDOT:PSS substrate at 4000 rpm for 40 seconds followed by annealing process at 110 °C for 10 min in a nitrogen-filled glovebox. Then 2PACz in ethyl alcohol (0.02 mg mL^{-1}) is spin-coated on the PVK:PBD substrate at 5000 rpm for 30 s followed by annealing process at 80 °C for 10 min in order to increase interface hydrophilicity. Perovskite is spin-coated at 3000 rpm for 60 s during which 240 μL of ethyl acetate is rapidly dispensed as an anti-solvent onto the film at 25 ~ 30 s window, and bakes at 60 °C for 5 min. 3TPYMB (40 nm) and LiF/Al electrodes (1 nm/100 nm) are then deposited in sequence, using a thermal evaporation system under a high vacuum of less than 10^{-4} Pa. The device active area is 4 mm^2 as defined by the overlapping area of the ITO and Al electrodes.

Characterization

The performance of the devices including EL spectra, *J*-*V*, *L*-*V*, *I*-*V*, and EQE-*J* characteristics are measured on a test system containing a QE Pro650 spectrometer, a source meter (Keithley 2400), and an integrating sphere (50 mm in diameter; Shenzhen Pynect). The as-fabricated device is placed in the sample chamber of the integrating sphere and get connected with power probe. The driving voltages is imposed from 0 V (with a step of 0.25 V) until the device gets burnt down, and the data spot is collected with a fix integration time of 100 ms. All the above tests are performed on device without encapsulation and carried out in a nitrogen-filled glovebox at room temperature. The operational stability of unencapsulated PeLEDs is evaluated by using a commercial OLED aging system (ZJZCL-1) in a nitrogen-filled glovebox. The capacitance of the devices is measured by the TH2829X series automatic transformer test system. For time-dependent microscopic images, the samples with a configuration of glass/perovskite/Ag are fabricated, where two parallel Ag electrodes with a spacing of 200 μm is deposited on the perovskite films. The custom-built measurement setup contains a commercial optical microscope with integrated lighting source (Nikon ECLIPSE L200N), and a source meter unit (KUAIQU SPS-E305). A constant bias of 6.0 V is imposed on the samples and the microscope images are collected at a 60-s interval at the ambient environment. TRPL data is performed by using a time-correlated single photon counting measurement with an excitation of a picosecond laser (405 nm, 4 MHz). Optical absorption spectra are collected by UV-vis spectroscopy (Perkin Elmer, Lambda 950). For the temperature-dependent PL, the sample is put into a vacuum cryostat (MicrostatN_2, Oxford Instruments) with controlled digital temperature from 100 to 300 K. A 405 nm-CW laser is adopted to avoid the amplification effect. The TA spectra experiments are performed by using commercial equipment (Helios, Ultrafast system). Absorption spectra are measured on a Perkin Elmer Lambda 950 UV-vis-NIR spectrometer. The fundamental excitation source is split into two parts. One part is used to generate a pump pulse at 400 nm via pumping the optical parametric amplifier. Another part is focused on the CaF_2 to generate a white light continuum spectrum for probe pulses. XPS are analyzed on a Thermo Scientific K-Alpha. FTIR data are tested using Thermo Fisher Scientific Nicolet iS5. Solid-state ^{207}Pb NMR spectra are collected by 300 MHz NMR spectrometer (Bruker Avance III). XRD measurements are conducted using Cu Kα radiation over an angular range from 10° to 60° with a scanning speed of 3° min^{-1} (Bruker D8). The spectra are recorded using a single-pulse excitation pulse sequence with a recycle delay of 1 s. SEM is characterized by the

Sigma300 microscope. AFM is characterized by Bruker Dimension Icon. Kelvin probe force microscopy (KPFM) measurements were performed using a Veeco Multimode atomic force microscope to determine the contact potential difference. Steady-state PL spectra and PLQYs are conducted on Edinburgh Instruments FS5 spectrofluorometer, which is equipped with an integrating sphere and uses a 150 W xenon lamp as the excitation source. In situ PL spectra are collected by using a spectrometer (QE Pro). UPS data are collected via Thermo Fisher Scientific ESCALAB XI+. The surface temperatures of PeLEDs are monitored by using an infrared thermal camera (FLIR T630sc) in ambient air. The Hall mobilities of perovskite films are measured on an Ecopia HMS-7000 Hall effect measurement system, with a calibrated blue LED ($\lambda = 465$ nm and maximum power 20 W) as the excitation source ensuring a high signal-to-noise ratio. Perovskite films are solution-grown on a fused silica substrate, and four-probe Au contacts are deposited in a desired Hall bar geometry. Photo-excitation is generated by illumination from the blue LED. The film conductivity is characterized with a Keithley 4200 analyzer and a LakeShore TTP4 probe station. The device with a structure of ITO/perovskite/Al (100 nm) is fabricated by thermal evaporation, employing a shadow-mask to define the channel width (1000 μm) and length (50 μm). The conductivity is calculated using the formula: $\sigma = l/AR$, where l, A and R represent the channel length, cross-sectional area, and resistance, respectively. We also measure the capacitance of perovskite films by using the Wayne Kerr 6500B device (with a collection frequency of 1 kHz).

Data availability

The data that support the plots within this paper and the other findings of this study are available from the corresponding authors upon request. The data supporting the findings generated in this study have been deposited in the Figshare database and are accessible via the link https://doi.org/10.6084/m9.figshare.30772967.46[46].

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Author contributions

X.Y. and L.W. conceived the idea and guided the work. X.Z., and Z.L. contributed equally to this work. X.Z. fabricated and characterized the films and PeLED devices. J.L. conducted the TRPL tests of perovskite films. Q.L. conducted the TA tests and Z.L. analyzed the data under the guidance of J.D.; J.W. taken microscope photographs of perovskite films. C.L. conducted the temperature-dependent PL tests. L.K. performed the Kelvin probe force microscopy. S.W. helped collect the data. R.D., Y.L. and H.S. contributed to the discussion of the results. X.Z. wrote the manuscript, which was revised by L.W., J.D. and X.Y. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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